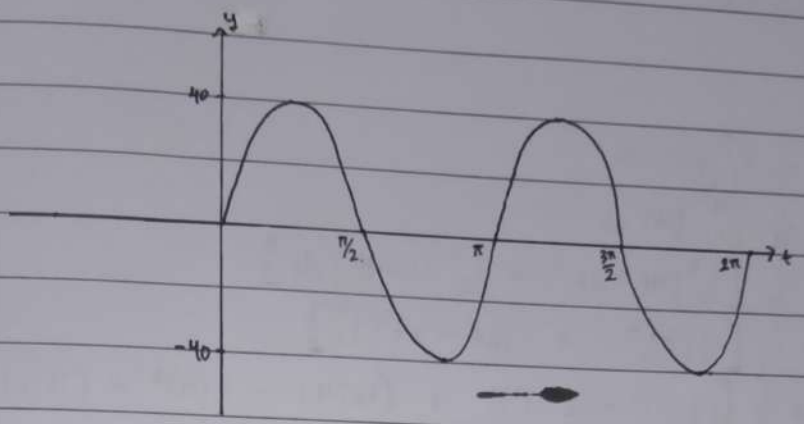
Date:

1. Diketahui bahwa $Y = i(t) = 40 \sin 2t$

Tentukan :

1. Grafik i(t)



2. iav

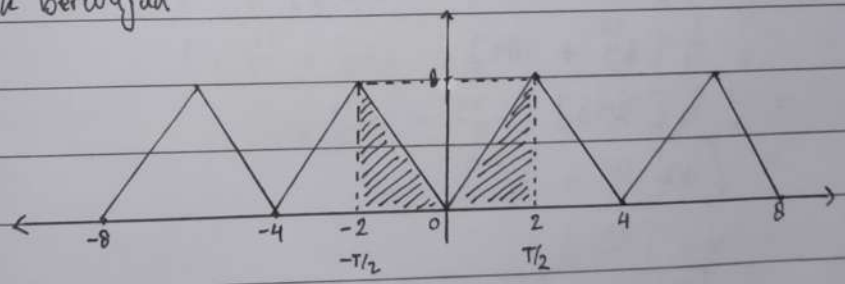
$$iav = \frac{1}{T}\int_0^T i(t)\,dt$$
$$= \frac{1}{\pi}\int_0^{\pi} 40 \sin 2t\,dt = 0$$

3. Irms

$$irms = \sqrt{\frac{1}{T}\int_0^T (i(t))^2\,dt}$$
$$= \sqrt{\frac{1}{\pi}\int_0^{\pi} (40 \sin 2t)^2\,dt} = 0$$

Karena simetris terhadap pusat (0,0) maka fungsi ganjil (add function) adalah 0

2. Diketahui grafik berwujud



Tentukan

1. Persamaan f(t) untuk 1 periode

$f(t) = \frac{2A}{T}t$ untuk $0 \le t \le \frac{T}{2}$

$= \frac{2(8)}{4}t = 4t$ untuk $0 \le t \le 2$

$f(t) = 2A - \frac{2A}{T}t$ untuk $\frac{T}{2} \le t \le T$

$= 2(8) - \frac{2(8)}{4}t = 16 - 4t$ untuk $\frac{T}{2} \le t \le T$

$f(t)$ $\begin{cases} 4t \text{ untuk } 0 \le t \le 2 \\ 16-4t \text{ untuk } 2 \le t \le 4 \end{cases}$

2. $f_{av} = \frac{1}{T}\int_0^T f(t)\,dt$

$= \frac{1}{4}\int_0^4 f(t)\,dt$

$= \frac{1}{4}\left[\int_0^2 (4t)\,dt + \int_2^4 (16-4t)\,dt\right]$

$= \frac{1}{4}\left[2t^2\Big|_0^2 + 16t - 2t^2\Big|_2^4\right]$

$= \frac{1}{4}\left[(2(2)^2 - (2(0)^2)) + (16(4) - 2(4)^2 - (16(2) - 2(2)^2))\right]$

$= \frac{1}{4}\left[8 + (32-24)\right]$

$= \frac{1}{4}\left[8+8\right]$

$= \frac{1}{4}\left[16\right] = 4$

3. $f_{rms} = \sqrt{\frac{1}{T}\int_0^T f^2(t)\,dt}$

$= \sqrt{\frac{1}{4}\int_0^4 f^2(t)\,dt}$

$= \sqrt{\frac{1}{4}\left[\int_0^2 (4t)^2\,dt + \int_2^4 (16-4t)^2\,dt\right]}$

$= \sqrt{\frac{1}{4}\left[\int_0^2 (16t^2)\,dt + \int_2^4 (16t^2 - 128t + 256)\,dt\right]}$

$= \sqrt{\frac{1}{4}\left[\frac{16}{3}t^3\Big|_0^2 + (\frac{16}{3}t^3 - 64t^2 + 256t)\Big|_2^4\right]}$

$= \sqrt{\frac{1}{4}\left[(\frac{16}{3}(2)^3 - \frac{16}{3}(0)^3) + (\frac{16}{3}(4)^3 - 64(4)^2 + 256(4) - (\frac{16}{3}(2)^3 - 64(2)^2 + 256(2))\right]}$

$= \sqrt{\frac{1}{4}\left[(\frac{128}{3} - 0) + (\frac{1024}{3} - 1024 + 1024 - (\frac{128}{3} - 256 + 512))\right]}$

$= \sqrt{\frac{1}{4}\left[\frac{128}{3} + (\frac{1024}{3} - (\frac{128}{3} + 256)\right]}$

$= \sqrt{\frac{1}{4}\left[\frac{128}{3} + (\frac{1024}{3} - (\frac{128}{3} + \frac{768}{3})\right]}$

$= \sqrt{\frac{1}{4}\left[\frac{128}{3} + (\frac{1024}{3} - \frac{896}{3})\right.}$

$= \sqrt{\frac{1}{4}\left[\frac{128}{3} + \frac{128}{3}\right]}$

$= \sqrt{\frac{1}{4}\left[\frac{256}{3}\right]}$

$= \sqrt{\frac{256}{12}} = \sqrt{\frac{64}{3}} = \frac{8}{3}\sqrt{3}$

No.

Date:

3. Diketahui $e(t) = 10t$ pada $-\pi < t < \pi$ periode 2π.

$$e_{av} = \frac{1}{T}\int_0^T e(t)\,dt$$

$$= \frac{1}{2\pi}\int_0^{2\pi} 10t\,dt$$

$$= \frac{1}{2\pi}\left[5t^2\Big|_0^{2\pi}\right]$$

$$= \frac{1}{2\pi}\left[5(2\pi)^2 - 5(0)^2\right]$$

$$= \frac{1}{2\pi}\left[5\cdot 4\pi^2 - 0\right]$$

$$= \frac{1}{2\pi}\left[20\pi\cdot\pi\right]$$

$$= \frac{20\pi}{2} = 10\pi$$

$$e_{rms} = \sqrt{\frac{1}{T}\int_0^T e^2(t)\,dt}$$

$$= \sqrt{\frac{1}{2\pi}\int_0^{2\pi}(10t)^2\,dt}$$

$$= \sqrt{\frac{1}{2\pi}\int_0^{2\pi}100t^2\,dt}$$

$$= \sqrt{\frac{1}{2\pi}\left[\frac{100}{3}t^3\Big|_0^{2\pi}\right]}$$

$$= \sqrt{\frac{1}{2\pi}\left[\frac{100}{3}(2\pi)^3 - \left(\frac{100}{3}(0)^3\right)\right]}$$

$$= \sqrt{\frac{1}{2\pi}\left[\frac{100}{3}(8\pi^3) - 0\right]}$$

$$= \sqrt{\frac{1}{2\pi}\left[\frac{800}{3}\pi^3\right]}$$

$$= \sqrt{\frac{800}{6}\pi^2}$$

$$= \sqrt{\frac{400}{3}}\,\pi$$

$$= \frac{20}{3}\sqrt{3}\,\pi$$